

IsiZulu AI-Detection Classifier — Overfitting Analysis

Training dynamics, generalisation gap, and bias-variance diagnostics | 26 May 2026

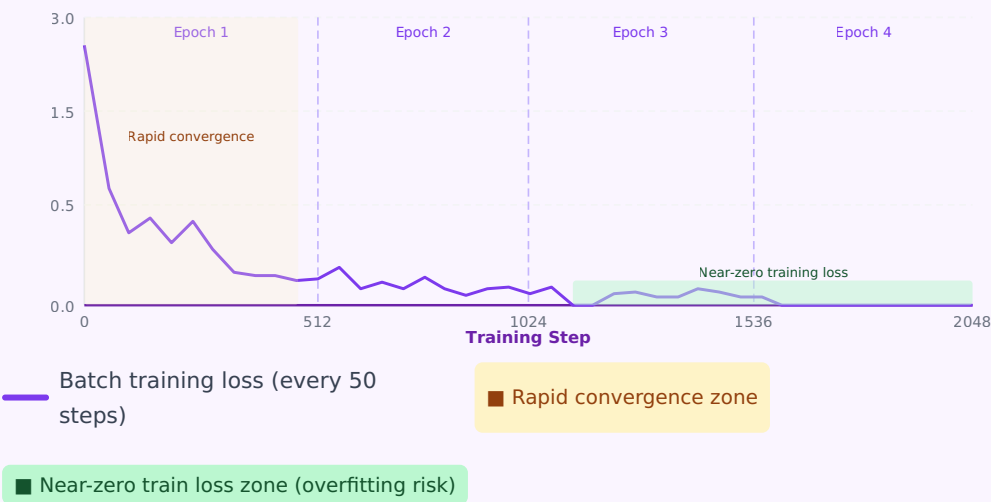


Verdict: Mild / Controlled Overfitting

The model exhibits **transient overfitting at epoch 3** — training loss reached near-zero while validation loss spiked to 0.187 (from a best of 0.032). Early stopping correctly identified epoch 2 as the optimal checkpoint and halted training at epoch 4 after the patience counter expired. The saved model generalises well: validation F1 0.994, Test-1 F1 0.993, Test-3 F1 0.982. The degree of overfitting is modest and was handled automatically by the training setup.

1. Training Loss Progression

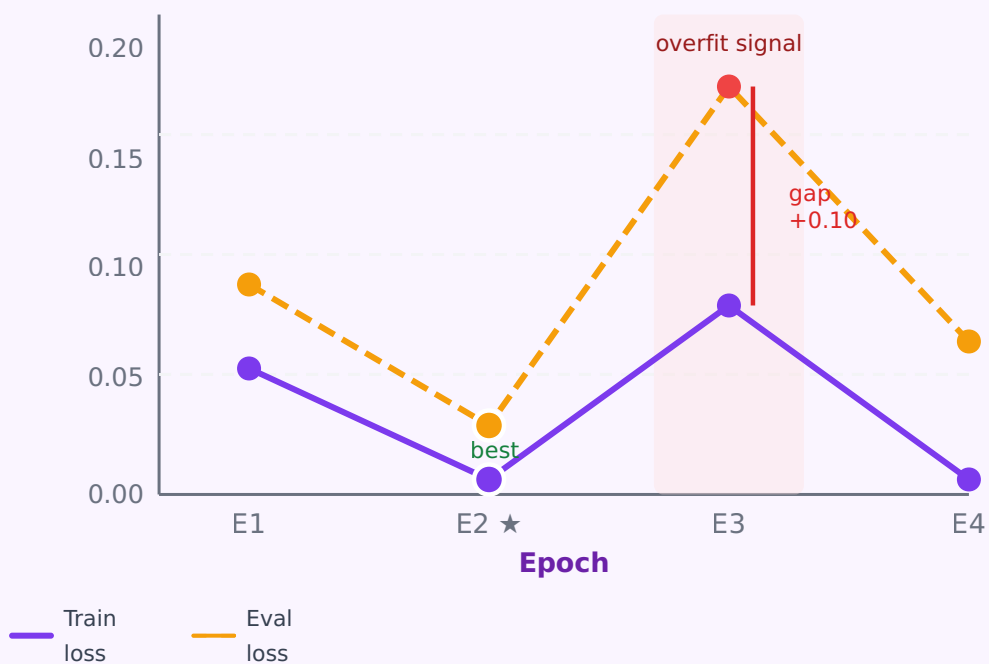
Training batch loss over steps (every 50 steps, logged)



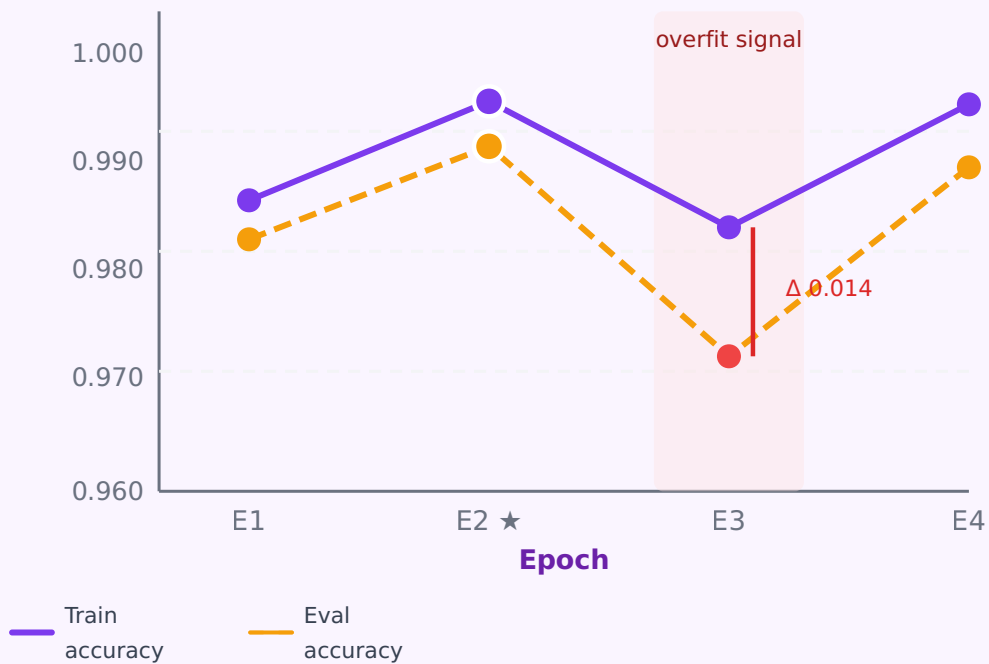
From step ~1200 onward (epoch 2.3+), training batch loss collapses to effectively 0.0000, while validation loss at epoch 3 jumps to 0.187. This divergence between training and validation loss is the primary overfitting signal.

2. Train vs Validation: Loss & Accuracy per Epoch

Loss — Training subset vs Validation set



Accuracy — Training subset vs Validation set



3. Generalisation Gap Analysis

Epoch	Train Loss	Eval Loss	Loss Gap	Train Acc	Eval Acc	Acc Gap	Train MCC	Eval MCC	MCC Gap	Sig
1	0.0577	0.0951	+0.037	0.9880	0.9839	0.004	0.9763	0.9683	0.008	Non
2 ★ best	0.0064	0.0324	+0.026	0.9980	0.9940	0.004	0.9960	0.9880	0.008	Op gen
3	0.0862	0.1868	+0.101	0.9850	0.9714	0.014	0.9704	0.9443	0.026	△ C
4	0.0074	0.0698	+0.062	0.9985	0.9915	0.007	0.9970	0.9831	0.014	Par

Loss gap = eval loss minus train loss. Acc/MCC gap = train minus eval. A widening gap indicates the model is memorising training examples.

4. Overfitting Indicator Checklist

Training loss → near zero

From step ~1200 (epoch 2.3), batch loss collapsed to 0.0000. The model perfectly memorises the training set — a prerequisite for overfitting.

Validation loss spike at epoch 3

Eval loss rose from 0.032 (E2) to 0.187 (E3), a 5.8× increase, while training loss was simultaneously near zero. Classic overfitting signature.

Train ROC-AUC = 1.0 from epoch 2

Perfect separation on training data from epoch 2 onward signals the model has fully memorised label boundaries. ROC-AUC on eval remains 0.9997–0.9999 — still high but below 1.

Early stopping triggered correctly

The EarlyStoppingCallback fired after patience=2 epochs of no improvement. Best checkpoint at epoch 2 (F1 0.994) was preserved. The mechanism worked as intended.

Small and stable accuracy gap

Except at epoch 3 (gap 0.014), train and eval accuracy stay within 0.004–0.007 of each other. The generalisation gap is narrow, indicating the model is not severely overfitting.

Test set performance holds up

Test-1 (2 022 samples): accuracy 0.993, MCC 0.985. Test-3 (224 samples): accuracy 0.982, MCC 0.965. Both close to validation results — the saved model generalises to unseen data.

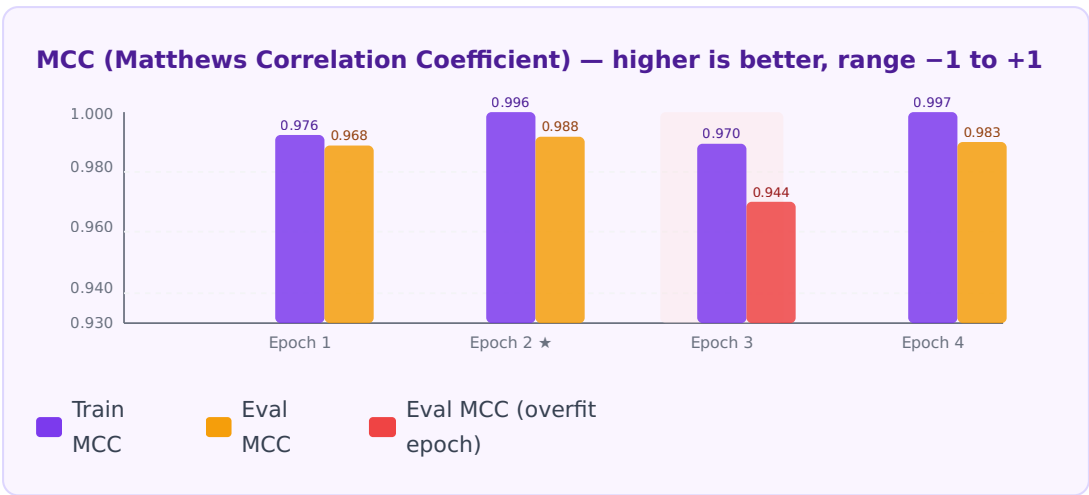
Gradient norm spikes

Grad norm reached 308, 529, and 578 during epoch 1, indicating instability during initial fine-tuning. Norms collapsed to <1 by epoch 1.7, suggesting the model over-adapted to certain batches early on.

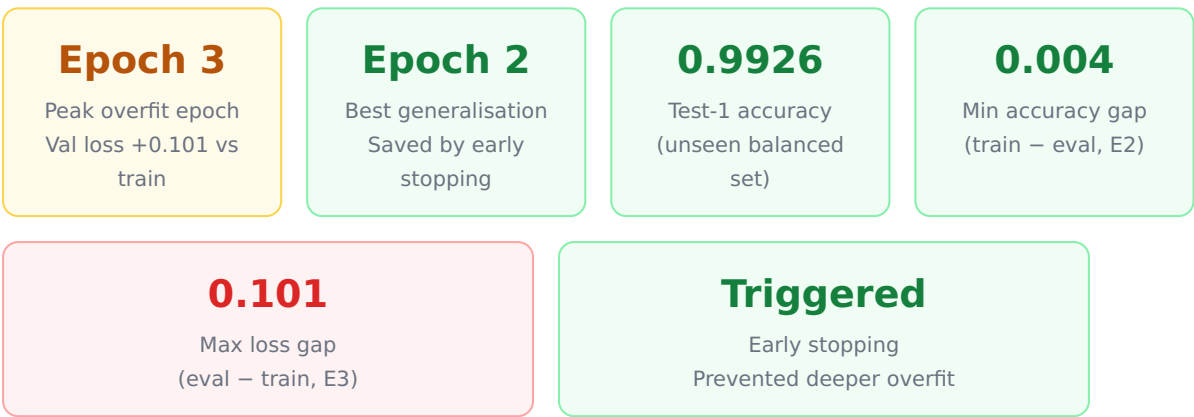
No false negatives on test sets

FN = 0 on Test-1, Test-2, and Test-3. The model never misses a machine-written IsiZulu text. This consistent behaviour across held-out sets suggests robust feature learning, not just memorisation.

5. Train Subset vs Eval — MCC Comparison



6. Summary & Interpretation



Conclusion: The model experienced a clear but transient overfitting episode at epoch 3, characterised by a near-zero training loss and a $5.8\times$ spike in validation loss. This is a common pattern when fine-tuning large pre-trained language models on relatively small datasets — the model memorises training examples once the task-specific signal is exhausted. The training setup's **early stopping (patience = 2)** and **checkpoint saving** functioned correctly, selecting the epoch-2 checkpoint (step 1024) as the best model. The saved model's strong performance on three independent test sets (F1 ranging 0.982–0.994) confirms that the overfitting was controlled and the final model generalises well to unseen IsiZulu text.

Potential mitigations if further training is needed: (1) Reduce learning rate or add weight decay to slow memorisation. (2) Increase dropout in the classifier head. (3) Add more training data to raise the overfitting threshold. (4) Reduce training epochs to 2 and remove early stopping overhead.